

MA

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Software Engineering I: Web Quiz

Instructions: This quiz is open-notes, closed-computer.

Incorrect questions have been corrected
in pen.

localhost:5000/results?a=&b=1

Above is a URL from a GET request. Answer the following multiple-choice questions about the Python code handling this URL:

1. `request.values.get("a")` returns which of the following:

a. None

b. ""

c. "true"

d. [throws an error]

2. `request.values.get("b")` returns which of the following:

a. None

b. "1" (string)

c. 1 (int)

d. [throws an error]

3. `request.values.get("c")` returns which of the following:

a. None

b. ""

c. False

d. [throws an error]



6

4. Consider the following code (with line numbers to the left):

base.html.j2:

```
1 <!DOCTYPE html>
2 <html lang="en">
3   <head>
4     <title>
5       {% block title %} {% endblock %}
6     </title>
7   </head>
8   <body>
9     {% block body %} {% endblock %}
10  </body>
11 </html>
```

index.html.j2:

```
1 {% extends 'base.html.j2' %}
2
3 {% block title %}
4   <title>Quiz Title</title>
5 {% endblock %}
6
7
8 {% for i in range(3) %}
9   <p id="paragraph">Paragraph # {%i%}</p>
10 {% endfor %}
```

When you run the code, you notice the following errors. Explain how you would fix each error. You may find it helpful to reference the line numbers.

a. `jinja2.exceptions.TemplateSyntaxError: Encountered unknown tag 'i'.`

on line 9, replace the `{%i%}` tag with `{%i33}`,
correct tuple format of `{%i33}`

b. None of the paragraphs show up on the page (they don't even show in 'view page source').

Put the code we want to actually render
inside the body block, by surrounding lines 8-10 with
`{% block body %}` and `{% endblock %}`

c. HTML validation error: "Stray end tag title"

we put an extra `<title></title>` tag around the
title, the base template already has close, remove them from

d. HTML validation error: "Duplicate ID paragraph" either the base or the index (line 4)

we are using `id="paragraph"` when there's multiple paragraphs
it should be class. Switch to `class="paragraph"` line 9

For the remainder of the quiz, assume you have a database to represent a list of users and the items they have purchased. The tables are set up as follows:

quiz_users		
id	first_name	email
1	Amanda	amanda.oneal@lakesideschool.org
2	Amanda	amanda.darling@lakesideschool.org
3	Tom	tom.butler@lakesideschool.org

quiz_purchased_items			
id	user_id	item_name	price
1	1	Pink Sweater	25.5
2	2	Purple socks	7.55
3	1	Blue jeans	35.47
4	3	Purple socks	7.55

5. Write a MySQL query to list the emails of all the people who purchased "Purple socks". With the example rows provided, I should get out "amanda.darling@lakesideschool.org" and "tom.butler@lakesideschool.org". You ONLY need to write the MySQL query. For full credit, you should NOT write any Python, Jinja, HTML, or any other kind of code for this question. For full credit, you must correctly use a JOIN in your query.

SELECT email FROM quiz_users JOIN
quiz_purchased_items ON quiz_users.id = quiz_purchased_items.
user_id WHERE item_name = "Purple socks"



6. The following code represents a search box where the user can search for a name in the database (using the same database tables as question #5), and the results page shows all email addresses associated with that name. Fill in the blanks such that all pages work together correctly, including sanitizing inputs to prevent SQL injection. You may assume that all code above and below the provided code is reasonable – for example, that the Python imports are correct.

app.py:

```
@app.route('/')
def index():
    return render_template('index.html.j2')

@app.route('/process', methods=['POST'])
def results():
    firstName = request.values.get("first")

    cursor = mysql.connection.cursor()
    query = "SELECT email FROM quiz_users
    WHERE email = %s"
    queryVars = ["first"]

    cursor.execute(query, queryVars)
    rows = cursor.fetchall()
    return render_template('results.html.j2', emailRows=rows)
```

Should be
first_name
not email

Should
already be
a string...
might catch
None type.

index.html.j2:

```
<body>
  <form method="POST" action="/process">
    First name:
    <input type="text" name="first" id="fName">
    <input type="submit">
  </form>
</body>
```

Should be
just

(first_name,)

results.html.j2:

```
<body>
  Email(s) for the name you searched:
  {% for row in emailRows %}
    <p> {{ row['email'] }} </p>
  {% endfor %}
</body>
```

Extra
blank param
to scop
SQL inject



7. The code from question #6 does not include client-side validation. Which one of the following additions would be reasonable client-side validation?

- a. In Python, check if firstName is not None before executing the MySQL query.
- b. In Jinja, check that firstName is not None before looping through the emails.
- c. In HTML, make the first name input have type="required" instead of type="text"
- d. In HTML, make the first name input use the "required" attribute while maintaining type="text"

8. The code from question #6 does not include server-side validation. Which one of the following additions would be reasonable server-side validation?

- a. In Python, check if firstName is not None before executing the MySQL query.
- b. In Jinja, check that firstName is not None before looping through the emails.
- c. In HTML, make the first name input have type="required" instead of type="text"
- d. In HTML, make the first name input use the "required" attribute while maintaining type="text"

9. The code from question #6 includes a variable called rows. Which one of the following is a possible value for rows?

- a. {'email': 'tom.butler@lakesideschool.org'}
- b. ({'email': 'tom.butler@lakesideschool.org'}, {'first_name': 'Tom'})
- c. ({'email': 'tom.butler@lakesideschool.org'},)
- d. ({'first_name': 'Tom'})

first_name is included

because it's referenced in our query:

//SELECT email FROM quiz_users WHERE first_name = %s"